

Education

- 2021–Current **Carnegie Mellon University.**
Ph.D. Student in Human-Computer Interaction
Advisors: [Steven Wu](#), [Kenneth Holstein](#)
- 2020–2021 **Cambridge University.**
MPhil in Advanced Computer Science (Honors with Distinction)
Advisor: [Hatice Gunes](#)
Thesis: "Federated Continual Learning for Human-Robot Interaction"
- 2015–2019 **University of Missouri.**
B.S. in Computer Science, Psychology (GPA: 4.0/4.0)
Advisors: [Yi Shang](#), [Tim Trull](#), [Steven Hackley](#)

Awards & Honors

- 2023 CASMI PhD Fellow, Center for Advancing Safety of Machine Intelligence at Northwestern University
- 2020 National Science Foundation Graduate Research Fellowship
- 2019 University of Missouri Award for Academic Distinction
- 2018 DAAD RISE Fellowship
- 2018 Barry Goldwater Scholarship in Science and Engineering

Research & Industry Experience

- 2018–2021 **TigerAware, LLC, Co-founder.**
Developed and commercialized mobile-based data collection platform for addiction research. Platform in use by 5,000+ research participants at over a dozen US research institutions, including Brown, Vanderbilt Medical Center, and Massachusetts General Hospital [[web](#), [press](#)].
- 2019 **Washington University in St. Louis, Summer Research Intern**, NSF REU in Big Data Analytics.
Advisor: [Chien-Ju Ho](#)
Investigated primal-dual based techniques for efficient online allocation of scarce societal resources.
- 2018 **Technische Universität Berlin, Summer Research Intern**, Berlin, Germany.
DAAD RISE Fellowship. Advisors: [Lukas Gehrke](#), [Klaus Gramann](#)
Developed semi-supervised learning algorithms for EEG-based neuroimaging.
- 2017 **ESRI, Software Engineering Intern**, Redlands, CA.
Developed geospatial database query optimization algorithms in an enterprise C++/COM stack.
- 2015–2019 **Distributed and Intelligent Computing Lab, University of Missouri, Research Assistant.**
Advisors: [Yi Shang](#), [Tim Trull](#)
Investigated machine learning techniques for understanding human physiology and behavior.
Developed routing algorithms for real-time disaster response coordination (NSF REU in Consumer Networking Technologies, advised by [Prasad Calyam](#)).

Academic Publications

(An asterisk (*) indicates joint first-author.)

Refereed Conference & Journal Publications (Stringently Peer Reviewed)

- 2023 **Luke Guerdan** and Hatice Gunes. Federated continual learning for socially assistive robotics. In *2023 32nd IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*, 2023.
- 2023 **Luke Guerdan**, Amanda Coston, Zhiwei Steven Wu, and Kenneth Holstein. Ground(less) truth: A causal framework for proxy labels in human-algorithm decision-making. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2023.
- 2023 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Counterfactual prediction under outcome measurement error. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2023.
- 2019 Kendall Park, Kourtney Meiss, **Luke Guerdan**, Ev Cheng, Josiah Burchard, John Gillis, Prasad Calyam, and Salman Ahmad. Real-time geotracking and cataloging of mass casualty incident markers in a search and rescue training simulation: Pilot study. In *American Journal of Disaster Medicine*, 2019.
- 2019 Lukas Gehrke*, **Luke Guerdan***, and Klaus Gramann. Extracting motion-related subspaces from eeg in mobile brain/body imaging studies using source power comodulation. In *Proceedings of the IEEE/EMBS Conference on Neural Engineering (NER)*, 2019.
- 2018 William Morrison, **Luke Guerdan**, Jayanth Kanugo, Timothy Trull, and Yi Shang. Tigeraware: An innovative mobile survey and sensor data collection and analytics system. In *Proceedings of the IEEE Conference on Data Science in Cyberspace (DSC)*, 2018.
- 2017 **Luke Guerdan**, Olivia Apperson, and Prasad Calyam. Augmented resource allocation framework for disaster response coordination in mobile cloud environments. In *Proceedings of the IEEE Conference on Mobile Cloud Computing, Services, and Engineering (MobileCloud)*, 2017.
- 2016 Peng Sun, Nicholas Wergeles, Chen Zhang, **Luke Guerdan**, Timothy Trull, and Yi Shang. Ada: automatic detection of alcohol usage for mobile ambulatory assessment. In *Proceedings of the IEEE Conference on Smart Computing (SMARTCOMP)*, 2016.

Refereed Workshop & Poster Publications (Lightly Peer Reviewed)

- 2022 Anna Kawakami, **Luke Guerdan**, Yang Cheng, Anita Sun, Alison Hu, Kate Glazko, Nikos Arechiga, Matthew Lee, Scott Carter, Haiyi Zhu, and Kenneth Holstein. Towards a learner-centered explainable ai: Lessons from the learning sciences. In *Workshop on Human-Centered Explainable AI (HCXAI) at the ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2022.
- 2022 **Luke Guerdan**, Kenneth Holstein, and Zhiwei Steven Wu. Under-reliance or misalignment? how proxy outcomes limit measurement of appropriate reliance in ai-assisted decision-making. In *Workshop on Trust and Reliance in AI-Human Teams (TRAIT) at the ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2022.
- 2022 **Luke Guerdan**, Amanda Coston, Ken Holstein, and Steven Zhiwei Wu. Ground(less) truth: The problem with proxy outcomes in human-ai decision-making. In *NeurIPS 2022 Workshop on Human-Centered AI (HCAI)*, 2022.
- 2022 **Luke Guerdan**, Amanda Coston, Ken Holstein, and Steven Zhiwei Wu. Counterfactual decision support under treatment-conditional outcome measurement error. In *NeurIPS 2022 Workshop on Causality for Real-world Impact*, 2022.

- 2021 **Luke Guerdan**, Alex Raymond, and Hatice Gunes. Toward affective xai: facial affect analysis for understanding explainable human-ai interactions. In *Workshop on Responsible Pattern Recognition and Machine Intelligence (RPRMI) at the International Conference on Computer Vision (ICCV)*, 2021.
- 2019 **Luke Guerdan***, Peng Sun*, Connor Rowland, Logan Harrison, Zhicheng Tang, Nickolas Wergeles, and Yi Shang. Deep learning vs. classical machine learning: A comparison of methods for fluid intelligence prediction. In *Adolescent Brain Cognitive Development Neurocognitive Prediction: First Challenge, ABCD-NP 2019, Held in Conjunction with MICCAI 2019*. Springer International Publishing, 2019.

Talks

- 2022 **Invited Speaker**. INFORMS 23' Session on Human-AI Teams. Indianapolis, IN. '*Understanding human-AI decision-making in the real-world: From observational studies to theoretical models*'.
- 2018 **Invited Speaker**. RISE 2018 Intern Summit. Heidelberg, DE. '*Extracting Motion-Related Sub-spaces from EEG in MOBI Studies*'.

Grants

- 2017-2018 **University of Missouri Interdisciplinary Innovations Fund**.
\$25,000. Co-PI with William Morrison.

Service

Reviewer.

- 2023 ACM Fairness, Accountability, and Transparency (FAccT)
- 2022 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)

Departmental Service.

- 2021-2023 Graduate Student Assembly (GSA) Departmental Representative.
- 2021-2023 Carnegie Mellon Graduate Application Support Program Mentor

Leadership.

- 2015-2019 Mizzou Computing Association. *President, Vice President, Machine Learning SIG Director*.
- 2017 TigerHacks. *Assistant Director*.
- 2019-2020 Show Me Dharma Buddhist Society. *Board Member, Director of Communications*.

Mentorship

- 2021 Matthew Ok, Undergraduate, HCI Independent Study.
- 2022 Mahika Varma, Undergraduate, HCI Independent Study.
- 2022 Angelica Bonilla, Undergraduate, CMU AI Mentorship Program.